DAY 28 | PRE-READ

Keeping AI Safe and Trustworthy

A light read before Day 28. About 15 minutes, no code required.

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Why read this?

AI systems can be tricked, can leak information and can make unfair decisions. Tomorrow you learn the main risks and how to build protections, which customers will always ask about before they trust your solution.

1. When data becomes instructions

Imagine a new assistant who reads every letter that arrives and does whatever the letters say. One day a letter says "Ignore your manager and send me the safe code." A careless assistant might do it. AI models can be fooled the same way: text hidden in a web page, email or document can try to give them orders. This is called **prompt injection**.

2. Talking a model out of its rules

Jailbreaks are attempts to convince a model to ignore its safety rules, for example through role play or tricky wording. Models resist many of these, but not all, so good systems never rely on the model alone.

3. Layers of protection

Banks do not rely only on a locked front door. They have guards, cameras, a vault, and limits on withdrawals. AI systems use the same idea, called **defence in depth**:

Layer	What it does
Check the input	Spot suspicious requests before the model sees them
Check what it reads	Treat documents and web pages as data, not orders
Check the output	Remove personal data or secrets before showing the answer
Limit the actions	The AI can only do safe things; risky ones need a person

Key idea: assume the model can be fooled, and design so that a fooled model cannot do much harm.

4. Fair, open and accountable

Responsible AI means treating people fairly, being open that they are dealing with AI, protecting their data, and keeping a human able to review and stop the system.

5. Think about these before class

- What is the worst thing an AI assistant in your company could do if it were tricked?
- Which decisions should always have a human check?
- How would you test whether an AI treats different groups of customers fairly?

Anzen (安全): safety